Seminar: Open-exam questions

SOK-3081 | Chapter 5 and earlier material

Instructions

The questions below are selected from the **open sections of the 2022 and 2019 SOK-3020 Econometrics exams**. Questions and parts requiring material beyond Chapter 5 have been omitted. Original question wording is retained, apart from necessary units clarifications. Two defective answer lists in the 2019 exam have been corrected and are marked below. Fill-in fields are shown as bold **BLANK** labels.

Work through Exercise 1 together with the lecturer. Then work independently on Exercise 2. Use *S18_ch05_exam_seminar_student.R* in RStudio for calculations and answers. You do not need to work in Quarto.

For fill-in questions, choose from the options below the text. For multiple-choice questions, record the selected letter. Write code and comments for open-response questions.

Exercise 1: 2022 open exam

START UP

Write the START UP code for this part in the companion R script: begin with a clean workspace, load the required packages, and load only the tuna data.

Information on Tuna model

The following R code loads the "tuna" data:

```
load(url("http://www.principlesofeconometrics.com/poe5/data/rdata/tuna.rdata"))
```

The *tuna* data have 52 weekly observations.

Variable	Definition
sal1	Number of cans of brand #1 sold during week
apr1	Average price (in dollars) per can of brand #1 during week
apr2	Average price (in dollars) per can of brand #2 during week
apr3	Average price (in dollars) per can of brand #3 during week
disp	1 if store display for brand #1 during week but no newspaper ad
dispad	1 if store display for brand #1 during week and newspaper ad during week

Question 1

Estimate a linear regression model where sales of tuna of brand #1 is explained by the price of tuna of brand #1, #2 and #3, all measured in dollars, and dispad. In the subsequent interpretations, we round numbers to the nearest can.

Name the fitted model *tuna_model* and write your R code in the corresponding section of the script.

Question 2

Interpretation of slope of apr1

Fill in the text with options below:

In this model the slope of the own price variable is interpreted as follows. A **BLANK1** increase in the price of brand #1 would lead to a **BLANK2** of **BLANK3** cans sold of brand #1 per week.

Options		
one cent	one dollar	increase
decrease	25214	252
-25214	-252	

Interpretation of slope of apr2

Fill in the text with options below:

In this model the slope of the cross price variable of apr2 is interpreted as follows. A **BLANK4** increase in the price of brand #2 would lead to a **BLANK5** of **BLANK6** cans sold of brand #1 per week.

Options		
one cent	one dollar	increase
decrease	8020	80
-8020	-80	

Interpretation of slope of dispad

Fill in the text with options below:

In this model the slope of the dispad variable is interpreted as follows. Weeks with **BLANK7** of dispad increase the sales of brand #1 with **BLANK8** cans per week.

Options		
one cent	increase	decrease
11925	119	-11925
-119	presence	absense

Question 3

The R^2 value of the regression model is:

Option	Answer
A	0.779
B	0.950
C	0.997
D	0.879

Question 4

Find a 90% confidence interval for a 1 dollar increase in apr1. The alternatives indicate the lower and upper limits, rounded to two digits.

Option	Answer
A	(-34768.04, -15659.97)
B	(530.62, 15508.98)
C	(-804.68, 19426.96)
D	(9554.16, 14295.30)

Question 5

Interpretation of own price elasticity.

Calculate the own price elasticity for brand #1 at a price of 80 cents per can, with and without dispad. Keep the price of brand #2 and #3 at their mean levels.

Fill in the text with options below:

The elasticity with dispad is interpreted as follows. A 1% increase in the own price leads to a **BLANK9**% change in the sales of brand #1. The elasticity without dispad is interpreted as follows. A 1% increase in the price leads to a **BLANK10**% change in the sales of brand #1.

We find that the sales with dispad is equal to **BLANK11** cans per week, while the sales without dispad is equal to **BLANK12** cans per week. This makes the price elasticity with dispad more **BLANK13** compared to the elasticity without dispad. This **BLANK14** since the sales with dispad is **BLANK15** making the elasticity **BLANK16** inelastic.

Options		
-0.8	-1.25	-2.26
-4.79	-0.61	elastic
inelastic	makes sense	does not make sense
higher	lower	more
less	4214	16138
10531	11925	3618
0.8	1.25	2.26
4.79	0.61	

Question 6

Sales forecast

Brand #1 currenly sells at a price of 80 cents per can and dispad. The sales manager of brand #1 notices that brand #2 and #3 sells at a 10 cent discount compared to their mean prices. What is the effect on the sales of brand #1 of these price drops? We work with numbers rounded to the nearest can here.

Fill in the text with options below:

The baseline sales at a price of 80 cent, dispad and mean competing prices is **BLANK17** cans per week. When both competitors drop their price by 10 cents this **BLANK18** the sales with **BLANK19** cans per week. The change is approximately a **BLANK20** percent **BLANK21** in sales per week.

Options		
drop	gain	4214
increases	decreases	16138
1733	12371	3428
60.9	214.3	-26.4
12.8	10.74	3268
12160	-214.3	-60.9
-3214		

Question 7

Find the standard error of the change in sales calculated in the previous section. The information you have is that brand #1 currently sells at a price of 80 cents per can and dispad, and that brand #2 and #3 sells at a 10 cent discount compared to their mean prices. The standard error is:

Option	Answer
A	627.18
B	710.7
C	5670
D	4960

Question 8

Find a 95% confidence interval of the change in sales calculated in the two previous section. The information you have is that brand #1 currently sells at a price of 80 cents per can and dispad, and that brand #2 and #3 sells at a 10 cent discount compared to their mean prices. The alternatives indicate the lower and upper limits.

Option	Answer
A	(2446.4953, 25260.4530)
B	(7230.0736, 13227.3198)
C	(303.3495, 3162.8391)
D	(2143.0913, 22097.6685)
E	(-3126.0, -340.15)

Exercise 2: 2019 open exam

START UP

Write a new START UP for this part: clear the previous exercise's objects, load the required packages, and import only the asparagus data. Exercise 2 does not depend on Exercise 1. Load the data using the working link below:

```
asparagus <- read.table(
  "https://userpages.umbc.edu/~coates/ec611/Waugh.txt",
  header = TRUE, nrows = 200
)
names(asparagus) <- tolower(names(asparagus))
```

Read only the 200 data rows because the file contains an extra end-of-file marker. The last line makes the variable names lowercase to match the questions.

Information on Asparagus model

The data have 200 observations, each describing a bunch of asparagus. This is a cross-sectional quality-and-price example; no lagged variables or time-series tests are used.

In 1928, Frederick Waugh published a seminal paper where he measured the effects of quality on price of asparagus using "multiple correlation" analysis. To eliminate the effects of seasons and day-to-day changes in prices, Waugh used as a dependent variable the ratio of the actual price (P) of a bunch to the average market price (PM). The average market price, PM, was 2.782 dollars.

Variable	Definition
price	Relative price, $price = 100P/PM$, where $PM = 2.782$ dollars
green	Number of inches of green colour on the asparagus, measured in hundredths of inches
nostalks	Number of stalks per bunch
disperse	Variation in size per bunch, measured in percent; 0 means uniform or equal size

Use this working university-hosted source when writing the START UP for this part. Read the header and the first 200 data rows, then convert the column names to lowercase. The original exam's download link is no longer used.

Units clarification: The original exam describes the relative-price index as "percent". A change of one index point means one percent of the fixed benchmark PM; it is not automatically a 1% change from the bunch's initial price. In Question 10, the original "10 percent increase in disperse" means an increase of 10 percentage points in that variable.

Question 9

Estimate a multiple regression model with price as the dependent variable, using the other variables as explanatory variables. We round answers to two digits.

Name the fitted model *asparagus_model* and write your R code in the corresponding section of the script.

Question 10

Interpretation of slope of green

Interpret the slope of green, when the change in green is 2 inches. Remember green is the number of inches of green color on the asparagus (measured in hundredths of inches).

Fill in the text with options below:

When green increase with **BLANK22** the relative price of asparagus **BLANK23** with **BLANK24** percent.

Options		
13.76	two inches	0.14
one hundredth of an inch	two hundredths of an inch	one inch
27.52	decrease	0.28
increase		

Interpretation of slope of nostalks

Interpret the slope of nostalks, when the number of stalks per bunch decrease with 5 stalks. Remember when nostalks, the number of stalks of asparagus per bunch increase, each asparagus in a bunch is smaller.

Fill in the text with options below:

In this model the slope of the nostalks variable is interpreted as follows. A **BLANK25** stalks decrease in the number of asparagus per bunch would **BLANK26** the relative price of asparagus with **BLANK27** percent.

Options		
6.79	1	decrease
-6.79	5	-1.36
1.36	increase	

Interpretation of slope of disperse

Interpret the slope of disperse, when it increases with 10 percent. Remember when disperse, the variation in size per bunch is 0, all asparagus are of equal size. It is measured in percent.

Fill in the text with options below:

In this model the slope of the disperse variable, when it is increased with 10 percent, it is interpreted as follows. A 10 percent increase in disperse leads to a **BLANK28** reduction in the relative price of asparagus.

Options		
percent	-3.45	unit
-0.35	0.35	3.45

Question 11

Find a 95% confidence interval for a 5 unit increase in nostalks.

The alternatives indicate the lower and upper limits, rounded to two digits.

Option	Answer
A	(-8.27, -5.30)
B	(1.65, 1.06)
C	(-1.65, -1.06)
D	(8.27, 5.30)

Question 12

Predict the relative price change (in percent) when disperse=0 and disperse=50, keeping the effect of all other variables constant (at their mean values).

Fill in the text with options below:

The relative price when disperse=0 is **BLANK29** percent. When disperse=50, the relative price **BLANK30** to **BLANK31** percent. The change in relative price due to this change in disperse is **BLANK32** percent.

Options		
is unchanged	-26.4	34.28
12.371	1.216	77.97
drops	-60.9	60.9
32.68	10.74	-2.14
-3.214	12.8	17.26
increase	gains	95.23
2.14		

Question 13

Find the standard error of the change in the relative price when disperse=0 and disperse=50, keeping the effect of all other variables constant.

Answer options corrected from the original exam.

Option	Answer
A	15.67
B	17.26
C	0.13
D	6.48

Question 14

Find a 99% confidence interval for the change in the relative price when disperse=0 and disperse=50, keeping the effect of all other variables constant.

Answer options corrected from the original exam. Use the t distribution and define the change as disperse = 50 minus disperse = 0.

Option	Answer
A	(-33.96, -0.57)
B	(-29.97, -4.56)
C	(-34.13, -0.40)
D	(0.40, 34.13)

Question 15

Find a 95% prediction interval for the relative price when green=550, nostalks=20 and disperse=14. The three alternatives indicate the predicted value, the lower and upper limits respectively.

Option	Answer
A	(84.46, 53.76, 115.16)
B	(84.46, 58.74, 110.19)
C	(77.24, 46.52, 107.95)
D	(84.46, 82.2, 86.72)

Question 16

Effect of rescaling the dependent variable.

What is the effect in DOLLARS of a increase in nostalks of 10 stalks per bunch? Remember 100 cents is one dollar.

Fill in the text with options below:

When we increase nostalks with 10 stalks per bunch the average price (P) will **BLANK33** with **BLANK34** cents.

Options		
17	12.8	26.4
increases	10.74	38
decreases	123	60.9
60	21	32.68
21.43	32.14	16
34	12.16	